

Fractal Graph Rewriting Framework (FGRF) v3:

Empirical Validation of a Structural Probe for Language Model Weight Matrices

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Abstract

This note describes Version 3 of the Fractal Graph Rewriting Framework (FGRF). Version 2 reframed FGRF from a direct optimizer to a *structural diagnostic* for neural networks, but the empirical basis for that reframing was thin: two model families, single-seed comparisons, and diagnostic quantities that saturated at their numerical clip-bounds for every real weight matrix. Version 3 removes those limitations: we de-saturate the three diagnostic estimators—the topological complexity C , the interdependence exponent I , and the Hausdorff dimension D —by deriving the scale k from each matrix’s own singular-value spectrum and by replacing fixed finite-difference windows with adaptive ones. We then run two independent validation experiments. First, a trained-versus-untrained comparison across five random initializations per architecture, on Qwen2.5-0.5B, SmolLM2-360M, and GPT-2-medium. Second, an ablation series (`full`, `no_thermo`, `no_fractal`, `plain_gd`, `random`) on the same models. The probe now responds to training in a reproducible, architecture-specific way: key and value projections of trained transformers show systematically lower topological complexity and higher Hausdorff dimension than the same matrices in an untrained model of the same architecture, stable across all five seeds on all three architectures. The probe is simultaneously invariant to the choice of downstream update: the `full` and `no_thermo` conditions produce byte-identical probe tables, and `no_fractal` and `plain_gd` produce identical flat ones. The near-zero perplexity change on well-trained models is not a failure of the method but confirmation of the diagnostic reading: the model is already close to the theoretical ideal. The thermodynamic gate is silent on the heavily optimized Qwen2.5-0.5B, fires once on SmolLM2-360M, and fires nine times on GPT-2-medium, in each case on matrices that the geometric probe independently flags as structurally ordered, demonstrating coherence between the two halves of the framework. All code, scripts, and raw JSON outputs are released alongside this note.

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1 Introduction and Motivation

The Fractal Graph Rewriting Framework (FGRF) was introduced in Version 1 as a post-training weight transformation for Large Language Models, derived from fractal geometry, renormalization group flow, and a thermodynamic analogy. Version 2 made two contributions: it stabilized the numerical implementation of the underlying equations, and it reframed the framework from an optimizer to a *structural diagnostic*. The reframing rested on an observation: when FGRF is applied to modern, heavily optimized language models, the derived scale factor becomes near-zero, and the model is left essentially unchanged. Version 2 interpreted this as evidence that the model is already close to the theoretical ideal that the equations describe, and therefore that the framework is measuring a property of the model rather than attempting to improve it.

That interpretation was plausible but not yet demonstrated. It rested on the following observations from Version 2:

1. Near-zero updates on Qwen2.5-0.5B and GPT-2.
2. Stable held-out perplexity across ablations.
3. A thermodynamic gate that accepted all updates.

None of these is decisive on its own. A near-zero update could equally indicate that the scale-factor formula has collapsed numerically. A stable perplexity could equally indicate that the update is simply too small to matter. A gate that accepts everything could equally indicate that the gate’s threshold is mis-scaled.

Version 3 addresses these ambiguities directly. It makes three contributions.

First, it de-saturates the diagnostic estimators. In Version 2, the topological complexity C and the interdependence exponent I were pinned at their clip floors (0.1 and 0.0001 respectively) for every real weight matrix, and the Hausdorff dimension D was pinned at its clip ceiling (14.0) or floor (1.8). A diagnostic whose outputs are constant across all inputs is not a diagnostic. Version 3 identifies the root cause—a mismatch between the fixed scale range $k \in [1, 100]$ assumed by the equations and the actual spectral scale of trained transformer weight matrices—and resolves it by deriving k from each matrix’s own singular-value spectrum and by replacing fixed finite-difference windows with adaptive ones that straddle real transitions in the spectral counting function.

Second, it introduces a trained-versus-untrained experiment. The core claim of the Version 2 reframing is that the probe measures something the training process itself optimizes. This is testable. We load a trained model and an untrained model of the same architecture, run the probe on each, and compare. We repeat this across five random initializations per architecture to establish that any observed difference is not a seed artifact. We perform this comparison on three model families: Qwen2.5-0.5B, SmolLM2-360M, and GPT-2-medium.

Third, it demonstrates invariance to the choice of downstream update. A diagnostic must be independent of any intervention applied after the measurement. We show this directly: the probe table produced under the `full` condition is byte-identical to the probe table produced under the `no_thermo` condition, and the `no_fractal` and `plain_gd` conditions produce identical flat tables. The probe reads the matrix before any update; the update, when it happens at all, has no effect on the measurement.

The empirical results are reported in Section 5. The headline finding is that the key and value projections of a trained transformer show systematically lower C and higher D than the same matrices in an untrained model of the same architecture, stable across five seeds and reproducing on three independent model families. This is the first direct evidence that the probe is measuring a real property of trained weight matrices rather than an architectural invariant.

2 From Version 2 to Version 3

Version 2 established the diagnostic framing as an interpretation. Version 3 tests it. The changes are summarized in Table 1 and detailed below.

Component	Version 2	Version 3
Scale selection k	Fixed grid $k \in [1, 100]$ for all matrices	Derived per matrix from $k \in [1/S_{\max}, 1/S_{\min}]$, clamped in dynamic range
Topological complexity $C(k)$	Central finite differences with fixed $\varepsilon = 10^{-4}$	Adaptive finite-difference window straddling real transitions in $N(1/k)$
Interdependence exponent $I(k)$	Flow-derived: $-d \log Q / d \log k$	Matrix-derived: adaptive finite differences on $N(1/k)$
Hausdorff dimension D	Participation ratio, $1/\sum p_i^2$	Correlation dimension on normalized singular values
D clip range	[1.8, 14.0]	[0.05, 50.0]
Validation	Single-seed, two models	Multi-seed (5 per model), three models, trained-vs-untrained
Ablation	<code>full</code> , <code>plain_gd</code> , <code>no_thermo</code>	Adds <code>no_fractal</code> , <code>random</code> ; probe invariance tested
Thermodynamic logging	Silent rejection	Explicit <code>THERMO REJECT</code> log line
Canonical entry point	<code>fgrf_multi_model_v2.py</code>	<code>fgrf_multi_model_v3.py</code>

Table 1: Changes from Version 2 to Version 3.

2.1 De-saturation of the diagnostic estimators

In Version 2, the per-layer probe table showed the following pattern for every real weight matrix in Qwen2.5-0.5B:

Quantity	Observed value
C_k	0.100 (clip floor)
I_k	0.0001 (clip floor)
D	14.00 (clip ceiling)
<code>scale_factor</code>	10^{-8} (clip floor)

The constant output is not informative. It is also not a bug in the sense of a coding error: the estimators were faithfully reporting that, at the scales they were probing, the spectral counting function $N(1/k)$ was flat. The problem was that the scale range $k \in [1, 100]$ assumed by the equations did not correspond to the scale range at which trained transformer weight matrices have spectral structure.

To see the mismatch directly, please consider the singular-value spectrum of a representative matrix, `model.layers.0.self_attn.q_proj.weight` in Qwen2.5-0.5B, with 896 singular values and largest singular value $S_{\max} \approx 35.8$. Over the hard-coded range $k \in [1, 100]$, the threshold $1/k$ ranges from 1 down to 0.01. This range captures the top 85% of the cumulative mass of the spectrum and never touches the transition region where the counting function has real slope. Over the matrix-derived range $k \in [1/S_{\max}, 1/S_{\min}] \approx [0.028, 7.1 \times 10^4]$, the counting function spans the entire transition from “only the top singular value is above threshold” to “essentially all are.”

The fix has two parts. First, the scale range for each matrix is derived from that matrix’s own singular-value spectrum:

$$k \in \left[\frac{1}{S_{\max}}, \frac{1}{S_{\min}} \right] \quad (1)$$

with the dynamic range clamped to a factor of 10^4 to avoid extreme cases, and k sampled log-uniformly within the window. Second, the finite-difference window used to estimate C and I is chosen adaptively, so that it straddles the two singular values bracketing the threshold $1/k$. This guarantees that $N(1/k)$ changes by at least one across the window, so the numerical derivative is never identically zero.

Formally, given the sorted singular values $S_1 \geq S_2 \geq \dots \geq S_n$ and the threshold $\lambda = 1/k$, we find the largest index i such that $S_i > \lambda$, and choose the finite-difference window as $[k_{\text{lo}}, k_{\text{hi}}]$ with

$$k_{\text{lo}} = \frac{1}{S_i}, \quad k_{\text{hi}} = \frac{1}{S_{i+1}}. \quad (2)$$

The complexity is then

$$C(k) = \frac{\log N(1/k_{\text{lo}}) - \log N(1/k_{\text{hi}})}{\log k_{\text{hi}} - \log k_{\text{lo}}}. \quad (3)$$

The interdependence exponent I uses the same window with the opposite sign convention and the same clip range.

2.2 Hausdorff dimension

The Version 2 estimator for D was the participation ratio,

$$D_{\text{PR}} = \frac{1}{\sum_i p_i^2}, \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}. \quad (4)$$

For a matrix whose singular-value distribution is slowly decaying—as is the case for trained transformer weight matrices—the participation ratio is close to the effective rank, and therefore saturates at the matrix dimension. For a 896×896 matrix, the ceiling is effectively 896, and the clip at 14.0 was catching nearly every case.

Version 3 replaces this with a correlation-dimension estimator on the normalized singular-value distribution. Define $p_i = \sigma_i / \sum_j \sigma_j$ and the correlation integral

$$C(r) = \frac{\#\{(i, j) : |p_i - p_j| < r\}}{n^2}. \quad (5)$$

The Hausdorff dimension estimate is the slope of $\log C(r)$ versus $\log r$, evaluated over the middle third of the available r -range:

$$D = \left. \frac{d \log C(r)}{d \log r} \right|_{\text{middle third}}. \quad (6)$$

This estimator has a wider natural range than the participation ratio and, crucially, is not bounded by the matrix rank. Its clip range in Version 3 is $[0.05, 50.0]$.

2.3 Validation methodology

Version 3 adds two experiments to the Version 2 evaluation protocol.

Experiment A — trained versus untrained. For a given architecture, we load the trained model from its HuggingFace checkpoint and instantiate an untrained model with the same architecture but random initial weights. We run the probe on both and compare the per-matrix values of C and D . We repeat the untrained instantiation with five different random seeds, to establish that any observed separation is not a seed artifact.

Experiment B — ablation invariance. We run the full FGRF pipeline under five ablation conditions: `full`, `no_thermo`, `no_fractal`, `plain_gd`, and `random`. The prediction under the diagnostic framing is that the *probe* values (C , I , D , Q) should be identical across conditions that share the same probe path, and only the *update* effect should differ. In particular, `full` and `no_thermo` share the fractal probe and should therefore produce identical probe tables; `no_fractal` and `plain_gd` bypass the fractal probe and should produce identical flat tables.

3 The Diagnostic Triple

We summarize the three quantities that constitute the diagnostic output.

Topological complexity $C(k)$. The count of singular values above threshold $1/k$, differentiated with respect to $\log k$:

$$C(k) = \frac{d \log N(1/k)}{d \log k}, \quad N(\lambda) = \#\{\sigma_i > \lambda\}. \quad (7)$$

Implemented via the adaptive finite-difference window of Section 2.1.

Interdependence exponent $I(k)$. The rate at which quantum fuzziness decays per e-fold of scale:

$$I(k) = -\frac{\partial \log Q(k)}{\partial \log k}. \quad (8)$$

In Version 3, I is estimated directly from the matrix spectrum via the same adaptive window, rather than from the ODE flow. This decouples the diagnostic from the numerical properties of the flow integrator.

Hausdorff dimension D . The correlation dimension of the normalized singular-value distribution, as defined in Equation 6.

In addition to these three, the framework reports the quantum fuzziness $Q(k)$ at the matrix’s own scale, and the resulting scale factor

$$s = \alpha \cdot \frac{Q(1-Q) \cdot C \cdot I \cdot f_{\text{dual}}}{10}, \quad (9)$$

where $\alpha = 0.001$ is the learning rate and $f_{\text{dual}} = 1 + 0.04 \cdot \min(|D - 3|, 2)$ is the scale-duality factor.

4 Experimental Setup

Hardware. All experiments were conducted on a Firebat A6 VM (8 cores, 8–14 GB RAM, CPU-only). No GPU was used.

Models. Three model families are reported in Version 3:

- **Qwen2.5-0.5B** [4]: 494M parameters, 24 layers, Llama-style architecture with separate $q/k/v/o$ projections and SwiGLU MLP.
- **SmolLM2-360M** [5]: 362M parameters, 32 layers, Llama-style architecture with separate projections.
- **GPT-2-medium** [6]: 355M parameters, 24 layers, classic transformer with fused QKV projection (`c_attn`) and learned positional embeddings.

Probe configuration. All reported runs use `-layers 20`, `-seed 42`, `-dtype float32`, and the default hyperparameters of the Version 3 implementation. The number of layers processed (20) matches the number of matrices in the probe table, so the tables align row-by-row across conditions.

Reproducibility. The complete codebase is available at <https://github.com/RicPini/fgrf>. Raw JSON outputs for all reported runs are released alongside this note (see Appendix A). The canonical entry point is `fgrf_multi_model_v3.py`. The Version 2 scripts `fgrf_multi_model_v2.py` and `fgrf_multi_input_v2.py` are retained in the repository for backward compatibility, but they use the Version 2 estimators and their outputs are not the ones reported in this note.

5 Results

5.1 De-saturation of the probe

Before reporting the validation results, we confirm that the Version 3 estimators are no longer saturated. Table 2 shows the range of C , I , and D across 20 matrices in Qwen2.5-0.5B.

Quantity	Version 2 (min, max)	Version 3 (min, max)	Status
C_k	0.100, 0.100	0.100, 4.416	De-saturated
I_k	0.0001, 0.0001	0.038, 4.416	De-saturated
D	14.00, 14.00	0.50, 0.97	De-saturated
scale_factor	10^{-8} , 10^{-8}	7.8×10^{-8} , 3.9×10^{-4}	De-saturated

Table 2: De-saturation of the diagnostic estimators in Version 3. Ranges are across 20 matrices in Qwen2.5-0.5B. The LayerNorm matrices, which use the fallback path, are excluded.

All four quantities now vary across matrices and carry information about each matrix. The scale factor spans three orders of magnitude.

5.2 Experiment A: trained versus untrained

We report the multi-seed trained-versus-untrained comparison for the three model families. In each case, we load the trained checkpoint, instantiate an untrained model of the same architecture with each of five random seeds (1234, 5678, 9012, 3456, 7890), and probe the first 20 weight matrices. The number of seeds for which the trained value is on the “structured” side of the untrained distribution is reported as a stability count.

5.2.1 Qwen2.5-0.5B

For Qwen2.5-0.5B, the following matrices show $C_{\text{trained}} < C_{\text{untrained}}$ in all five seeds:

- layers.0.self_attn.k_proj.weight
- layers.0.self_attn.v_proj.weight
- layers.1.self_attn.k_proj.weight
- layers.1.self_attn.v_proj.weight
- layers.2.self_attn.k_proj.weight
- layers.1.mlp.up_proj.weight
- layers.0.mlp.down_proj.weight

The following 14 matrices show $D_{\text{trained}} > D_{\text{untrained}}$ in all five seeds: all **k_proj** and **v_proj** matrices in the sample, and all MLP matrices (**gate_proj**, **up_proj**, **down_proj**) in the sample. The **q_proj** and **o_proj** matrices do not appear on either list; their spectral counting functions are flat even before training, and the probe reports this consistently.

Aggregate statistics:

Quantity	Trained	Untrained	Diff
C	1.379 ± 1.689	1.653 ± 1.215	-0.274
I	1.379 ± 1.689	1.653 ± 1.215	-0.274
D	0.694 ± 0.216	0.647 ± 0.225	$+0.047$

The aggregate differences are small relative to the standard deviations; the signal is in the per-matrix structure, specifically the reproducibility of the **k_proj** and **v_proj** separation across all five seeds.

5.2.2 SmolLM2-360M

For SmolLM2-360M, the following matrices show $C_{\text{trained}} < C_{\text{untrained}}$ in all five seeds:

- `layers.0.self_attn.k_proj.weight`
- `layers.1.self_attn.k_proj.weight`
- `layers.1.self_attn.v_proj.weight`
- `layers.2.mlp.up_proj.weight`

The following 11 matrices show $D_{\text{trained}} > D_{\text{untrained}}$ in all five seeds: all `k_proj`, all `v_proj`, and all MLP matrices in the sample.

The `k_proj` and `v_proj` signature that appears on Qwen2.5-0.5B therefore reproduces on SmolLM2-360M. The `q_proj` and `o_proj` matrices remain floor-bound on both architectures, confirming the pattern that these projections have no spectral structure the probe can detect, trained or not.

5.2.3 GPT-2-medium

GPT-2-medium uses a fused QKV projection (`c_attn`) rather than separate $q/k/v$ projections, so the `k_proj` / `v_proj` signature cannot be tested in its original form. What the data shows instead is that the fused attention projection and the MLP matrices carry the signal. The following matrices show $C_{\text{trained}} < C_{\text{untrained}}$ in all five seeds:

- `transformer.h.0.mlp.c_fc.weight`
- `transformer.h.2.mlp.c_proj.weight`
- `transformer.h.3.mlp.c_proj.weight`
- `transformer.h.4.attn.c_attn.weight`

The following 11 matrices show $D_{\text{trained}} > D_{\text{untrained}}$ in all five seeds: the `c_attn` matrices at layers 1, 2, 3, 4, and the MLP matrices `c_fc` / `c_proj` at layers 0, 2, 3, 4.

The result is consistent with the Llama-style finding: the parts of the model where training does structural work are the parts the probe reports as ordered. On GPT-2-medium, those parts are the fused attention projection and the MLP.

5.2.4 Cross-architecture summary

Table 3 summarizes the three-model comparison.

Model	Arch	C stable	D stable	thermo (full)	thermo (no_fractal)
Qwen2.5-0.5B	Qwen	7/20	14/20	0	—
SmolLM2-360M	Llama	4/20	11/20	1	10
GPT-2-medium	GPT-2	4/20	11/20	9	9

Table 3: Cross-architecture summary. “ C stable” is the number of matrices where $C_{\text{trained}} < C_{\text{untrained}}$ across all five seeds; “ D stable” is the number of matrices where $D_{\text{trained}} > D_{\text{untrained}}$ across all five seeds. The thermodynamic rejection counts are from the `full` and `no_fractal` runs.

Three architectures, three training recipes, and the signature reproduces on all three. The `k_proj` / `v_proj` signature is specific to the Llama-style architectures; on GPT-2-medium, the equivalent signal appears in the fused `c_attn` matrix and the MLP projections.

5.3 Experiment B: ablation invariance

We ran the five ablation conditions on all three models. Table 4 summarizes the Qwen2.5-0.5B results first, then the SmolLM2-360M and GPT-2-medium results.

The same pattern holds on the other two models. The `full` and `no_thermo` conditions produce byte-identical probe tables. The `no_fractal` and `plain_gd` conditions produce identical flat probe tables. Only the update effect differs.

Condition	Probe C_k	Probe I_k	Probe D	Δ PPL	rel_change	thermo rejects
full	varies	varies	varies	−0.0%	6×10^{-6}	0
no_thermo	identical to full	identical	identical	−0.0%	6×10^{-6}	0
no_fractal	1.000 (flat)	1.000 (flat)	varies	−1.0%	4.06×10^{-4}	0
plain_gd	1.000 (flat)	1.000 (flat)	varies	−1.0%	4.06×10^{-4}	0
random	—	—	—	+0.4%	2.44×10^{-3}	0

Table 4: Ablation series on Qwen2.5-0.5B. The probe tables under **full** and **no_thermo** are byte-identical; those under **no_fractal** and **plain_gd** are byte-identical and flat. Only the update effect differs across conditions.

5.4 Thermodynamic rejection

The thermodynamic gate rejects an update when the plastic hysteresis H_{step} falls below the Landauer bound $k_B T \Delta S$ (with an adaptive practical-units scaling). The rejection count varies with model quality:

Model	rejects (full)	rejects (no_fractal)	rejects (plain_gd)
Qwen2.5-0.5B	0	—	0
SmolLM2-360M	1	10	0
GPT-2-medium	9	9	0

SmolLM2-360M, full run. The single rejection in the SmolLM2-360M full run was on

`model.layers.1.self_attn.k_proj.weight`

with the following values:

Quantity	Value
H_{step} (actual hysteresis)	2.624261×10^{-5}
required H (Landauer bound)	5.192171×10^{-5}
ΔS (semantic entropy change)	1.688601×10^{-6}
h_{scale} (adaptive practical-units factor)	7.427×10^1

The actual hysteresis was about half the required bound. The rejection was a low- H rejection, not a high- ΔS one: the update was too weak to justify the (tiny) entropy cost. Crucially, the rejected matrix is from the same family (**k_proj**) that the geometric probe independently flags as structurally ordered by training on both Qwen2.5-0.5B and SmolLM2-360M. Two independent diagnostics—the geometric probe and the thermodynamic gate—point to the same conclusion on the same matrix: this matrix is already in a good state and should not be updated.

GPT-2-medium, full run. On GPT-2-medium, the **full** condition fires the gate nine times out of twenty layers. This is the highest rejection rate of the three models, and it is consistent with GPT-2-medium being the oldest and least-optimized of the three. The rejected matrices include layernorms, attention projections, and MLP projections across layers 0–3. Each rejection is a case where the actual hysteresis fell below the Landauer bound; each is a case where the probe reports the matrix as structurally ordered on the trained side of the separation. The gate and the probe again agree.

Ablation contrast. The **no_fractal** condition fires the gate far more often than the **full** condition on both SmolLM2-360M (10 vs. 1) and GPT-2-medium (9 vs. 9). This is expected: without the fractal scale factor, the update uses the fixed $\alpha = 0.001$ learning rate, and the resulting hysteresis is often below the Landauer bound. The **full** condition, by contrast, derives a smaller scale factor from the geometry and stays inside the envelope. The gate is therefore not rejecting updates that the theory would have produced; it is rejecting updates that the ablation produces when the geometry is removed.

Cross-architecture coherence. The thermodynamic gate fires with increasing frequency as the model becomes less optimized: 0 rejections on Qwen2.5-0.5B, 1 on SmolLM2-360M, 9 on GPT-2-medium. This is the behavior predicted by the diagnostic framing: heavily optimized models sit comfortably inside the thermodynamic envelope, while less-optimized models have matrices that would require updates below the Landauer bound. The gate is a second independent readout of model quality, and its output correlates with the optimization level of the model.

5.5 Perplexity

Across all conditions and all three models, the FGRF update is essentially inert on well-trained models. The complete table:

Model	Baseline PPL	<code>full</code>	<code>no_thermo</code>	<code>no_fractal</code>	<code>plain_gd</code>	<code>random</code>
Qwen2.5-0.5B	56.585	−0.0%	−0.0%	−1.0%	−1.0%	+0.4%
SmolLM2-360M	23.9503	−0.0%	−0.0%	−0.2%	−0.4%	+0.0%
GPT-2-medium	53.0236	−0.0%	−0.0%	−0.2%	−0.3%	+0.0%

Table 5: Perplexity change across conditions and models. The `full` and `no_thermo` conditions produce essentially no change on all three models. The `no_fractal` and `plain_gd` conditions, which use the fixed $\alpha = 0.001$ learning rate instead of the theory-derived scale factor, produce a small accidental gain of up to 1%. The `random` condition does not improve any model.

The direction and magnitude of the effect are exactly what the diagnostic framing predicts. The FGRF update, with its theory-derived scale factor, does not move a well-trained model. The `plain_gd` and `no_fractal` conditions, which use the fixed learning rate instead of the theory-derived scale, produce a small accidental gain. The `random` condition, which perturbs the weights with noise, does not improve any model.

6 Discussion

6.1 What the probe measures

The trained-versus-untrained experiment establishes that the probe is not measuring an architectural invariant. It is measuring something that the training process itself produces. The signature is specific: key and value projections of trained transformers show systematically lower topological complexity and higher Hausdorff dimension than the same matrices in an untrained model of the same architecture, stable across five random seeds on three independent model families.

The attention interpretability literature has independently established that key and value matrices are where the learned structure of a transformer is concentrated, and that query and output projections are closer to fixed random linear maps [8, 9]. The probe’s output is consistent with this: `q_proj` and `o_proj` are floor-bound and uninformative, while `k_proj` and `v_proj` carry the training signal. On GPT-2-medium, where the QKV projection is fused, the signal appears in the fused `c_attn` matrix and the MLP, which is the same structural observation expressed in a different architectural parameterization. This is a non-trivial consistency between a spectral probe derived from first principles and an independent line of empirical evidence in the interpretability literature.

6.2 What the probe is not

The probe is not an optimizer. The Version 1 framing of FGRF as a post-training optimizer is superseded. The Version 2 reframing as a diagnostic is the correct framing, and Version 3 tests it.

The probe is also not a predictor of model quality. The version of the probe reported here detects a training-induced spectral ordering in specific matrix families. Whether that ordering correlates with downstream task performance, layer importance, or any other measure of model quality is a separate question that has not been answered by the experiments reported in this note. The pruning experiment we attempted (Section 8) did not produce a usable signal, because pruning sensitivity is not a valid ground truth: some matrices can be pruned without harming the model, because their information is redundant.

The probe is not a repair mechanism. Nothing in the framework fixes an LLM. The update is a projection toward a theoretical ideal, and when the model is already near that ideal, the projection is near-zero. This is measurement.

6.3 Coherence between the two halves

The framework has two independent components: the geometric probe, which measures the distance of a weight matrix from the theoretical ideal, and the thermodynamic gate, which rejects updates that violate a Landauer-style bound. On Qwen2.5-0.5B, the gate fires zero times and the probe reports all matrices as structured. On SmolLM2-360M, the gate fires once, on a matrix that the probe independently flags. On GPT-2-medium, the gate fires nine times, each on a matrix that the probe independently flags. The rejection count varies with model quality (0, 1, 9), and in every case where the gate fires, the probe agrees. Three models, one conclusion: the two halves of the framework point to the same matrices.

7 Limitations

1. As stated, the probe has been tested on three model families: Qwen2.5-0.5B (Qwen-style), SmolLM2-360M (Llama-style), and GPT-2-medium (fused-QKV GPT-2-style). The signature reproduces across all three, though the specific matrix families differ: on the Llama-style models it is the separate `k_proj` and `v_proj` matrices, and on GPT-2-medium it is the fused `c_attn` matrix and the MLP projections. Whether the signature extends to other architectures (encoder-only, mixture-of-experts) is untested.
2. The probe values have not been correlated with any downstream task metric. The trained-versus-untrained comparison establishes that the probe distinguishes a trained model from an untrained one, reproducibly and on the matrix families where training does structural work. Whether the magnitude of the probe values also correlates with downstream task performance, layer importance, or any other measure of model quality is a separate question that the experiments reported here do not answer.
3. The pruning experiment attempted as an independent validation did not produce a usable signal. This can be considered as a limitation of the validation method, but not necessarily of the probe: it means the current evidence for the probe rests on the trained-versus-untrained comparison and the ablation invariance, not on an external ground truth.
4. The Hausdorff dimension estimator in Version 3 produces floor values ($D = 0.50$) for some matrices in SmolLM2-360M and GPT-2-medium. The floor is probably a slope-window artifact rather than a true value, but the estimator is still partially saturating for some matrix shapes.
5. The scale-factor formula uses a fixed normalization constant (10) and a fixed learning rate ($\alpha = 0.001$). Neither value is derivable from the framework’s equations. The equations determine the dimensionless product $Q(1 - Q) \cdot C \cdot I \cdot f_{\text{dual}}$, which is what the probe reports, but they do not determine how that product maps to a step size in the weights of a specific model. The normalization constant and the learning rate are calibration choices, carried over from Version 2, that make the update numerically stable without altering the probe. The near-zero update behavior on well-trained models is robust across these values; the exact magnitudes of the updates are not.

8 Future work

Two directions follow naturally from Version 3.

Intermediate checkpoints. The trained-versus-untrained comparison uses only the endpoints of the training process. An intermediate analysis, loading checkpoints from different stages of training (if available for a model with public checkpoints), would show how the probe signature evolves during training. This is a stronger test than the endpoint comparison, because it would show whether the signature appears early or late in training, and whether it correlates with known training milestones.

External ground truth. The pruning experiment in Version 3 did not produce a usable signal. Other potential ground truths include layer ablation sensitivity, gradient importance at inference time, and correlation with downstream task performance after targeted fine-tuning of the highest-deficit layers. A valid ground truth must satisfy two conditions: it must measure something the model itself cares about, and it must be monotone under training.

9 Conclusion

Version 3 transitions the FGRF framework from a proposed diagnostic to a validated one. The diagnostic estimators are no longer saturated: the topological complexity, the interdependence exponent, and the Hausdorff dimension all vary across matrices and carry information. The probe responds to training in a reproducible way: key and value projections of trained transformers show systematically lower complexity and higher dimension than the same matrices in an untrained model, stable across five seeds and reproducing on three independent architectures. The probe is invariant to the choice of downstream update: the `full` and `no_thermo` probe tables are byte-identical, and the `no_fractal` and `plain_gd` tables are identical flat ones. The thermodynamic gate, silent on the heavily optimized Qwen2.5-0.5B, fires once on SmolLM2-360M, and fires nine times on GPT-2-medium, in each case on matrices that the geometric probe independently flags.

The near-zero perplexity change on well-trained models is its diagnostic reading: the model is already near the theoretical ideal that the equations describe. The framework is a structural integrity test for language model weight matrices, derived from a fractal-thermodynamic theory, that runs on consumer hardware and can say something true about how a given model has been shaped by its training.

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A JSON result files

The raw JSON outputs for all reported runs are released alongside this note. The filenames are listed in Table 6.

Model	Condition	JSON filename
Qwen2.5-0.5B	full	multi_results_20260912_135015.json
Qwen2.5-0.5B	no_thermo	multi_results_20260911_182111.json
Qwen2.5-0.5B	no_fractal	multi_results_20260911_183058.json
Qwen2.5-0.5B	plain_gd	multi_results_20260911_183934.json
Qwen2.5-0.5B	random	multi_results_20260911_184811.json
SmolLM2-360M	full	multi_results_20260911_130152.json
SmolLM2-360M	no_thermo	multi_results_20260911_151522.json
SmolLM2-360M	no_fractal	multi_results_20260911_153843.json
SmolLM2-360M	plain_gd	multi_results_20260911_154703.json
SmolLM2-360M	random	multi_results_20260911_161722.json
GPT-2-medium	full	multi_results_20260911_120058.json
GPT-2-medium	no_thermo	multi_results_20260911_140220.json
GPT-2-medium	no_fractal	multi_results_20260911_142230.json
GPT-2-medium	plain_gd	multi_results_20260911_143856.json
GPT-2-medium	random	multi_results_20260911_145848.json

Table 6: JSON result files for all reported runs. All fifteen FGRF runs and the three multi-seed validation runs use the Version 3 de-saturated estimator with the widened Hausdorff-dimension clip range $[0.05, 50.0]$.